



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df006-2017-android
Operator:	Floris
Analysis Started:	2026-06-18 11:04:08
Analysis Completed:	2026-06-18 11:11:52
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df006\2017\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df006\2017\dfdof_output\ft-df006-2017-android
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df006_external.001	drone_sd	physical	14.9 GB	2026-06-18 11:05:03
DF006_flight.zip	controller_android	logical	445.3 MB	2026-06-18 11:05:04
df006_internal.001	drone_flight_storage	physical	7.4 GB	2026-06-18 11:05:27

Evidence Source Path

C:\Users\Floris\Documents\ddji_drones\test_phantom_4\df006\2017\android

Cryptographic Hashes

df006_external.001		drone_sd	physical
SHA-256:	004071f5ee6d64cb85a6632b95720a03f0b662a2b1c565a1d8fa0b6249df3ab7		
SHA-1:	e1e054bec2474156b046e0e68e4ed31465176ef1		
DF006_flight.zip		controller_android	logical
SHA-256:	ef1c5b1dfc461c36ae08987ea2a0f281a11a79d280b595c987ab167e6304f8c0		
SHA-1:	22c0b1980a2f48ffe4096ea68405ab6613b94a86		
df006_internal.001		drone_flight_storage	physical
SHA-256:	53cb72f413544274315a6d468969802a4da401a4a2f342b78ea5cec4c07e0366		
SHA-1:	bd6eb1c7396b606e494fd144eded4cfcf7d5e853		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for DF006_flight.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for DF006_flight.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller ios]: source evidence not found
7. [p4 - controller ios]: source evidence not found
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY010.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	11	11	[OK]
exiftool	32	32	[OK]
fls	2	2	[OK]
icat	40	40	[OK]
mmls	2	2	[OK]
parser_android	1	1	[OK]
txtlogtcsvtool	2	2	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	3 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	1 / 7
Tools Succeeded	7 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Drone Serial Number	07JDE580020139	[K] Controller-only	2
Drone Model	P4 Professional P4P	[!] Inconsistent	8
DJI Application Version	4.1.2	[K] Controller-only	2
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	24	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Controller Serial Number	0BDDE5B001008U	[K] Controller-only	1
Last Known Latitude	39.961189998726496	[K] Controller-only	1
Last Known Longitude	-106.21651012463371	[K] Controller-only	1
Account UID	879825329191059456	[K] Controller-only	2
Account Nickname	dronefo...	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: acaa4c8dc5826e59f451bbbd7f40c9d2c4003b8e74e8709cc78835cefde155a5
Drone Log Name: FLY002.csv
Start: 2017-06-28T17:43:55.558000+00:00
End: 2017-06-28T17:54:29.423000+00:00
Duration: 633.9 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-06-28 17:50:23	drone_flight_storage:dl	Log started	P4P
2017-06-28 17:54:29	drone_flight_storage:dl	Log ended	

Correlated Artefacts

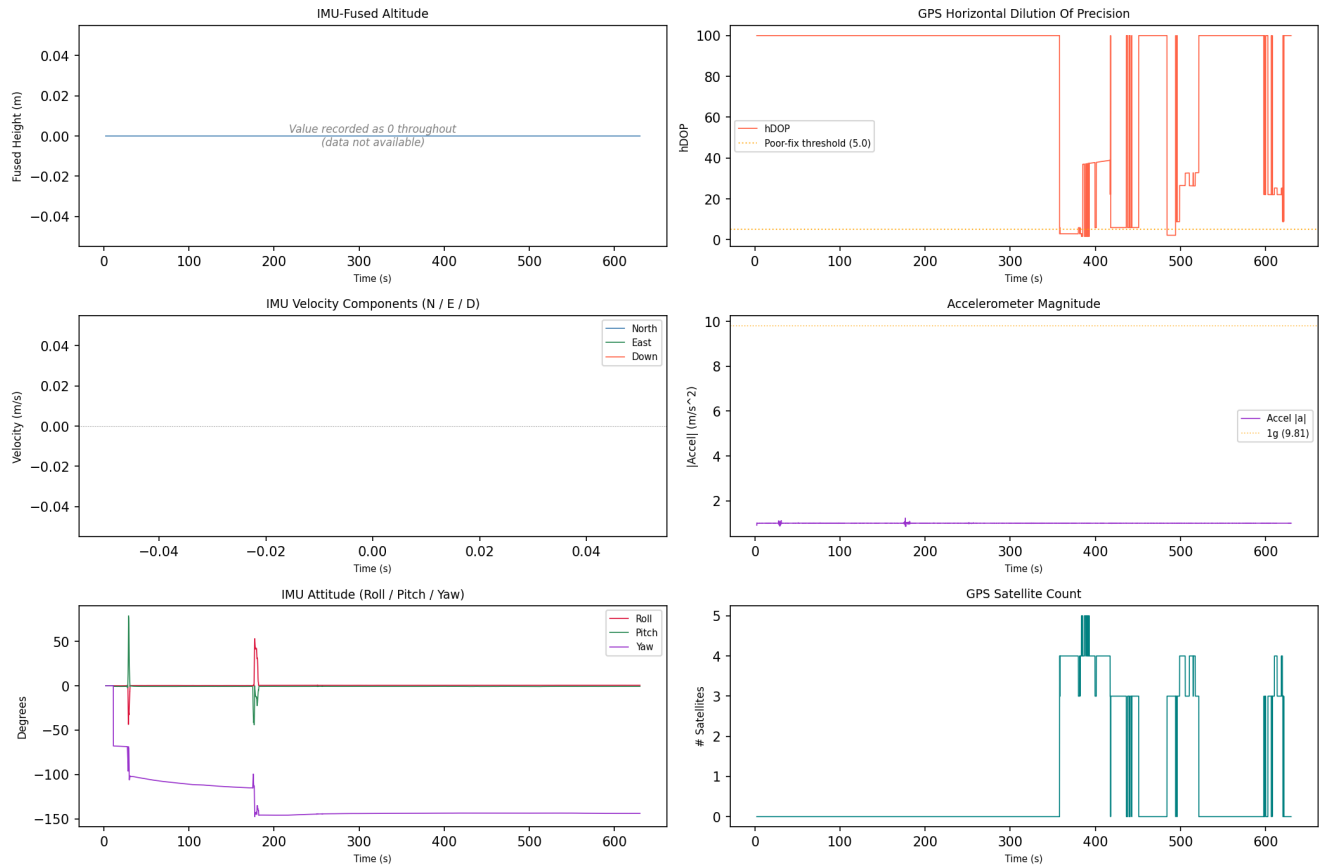
Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9574, lon -106.2177)
p5:9da5fd3a57eba8acf6c7c94462b7edb23eeb78287511457c684373c955f935ba	
drone_sd:images	GPS within flight bounding box (lat 39.9612, lon 106.2165)
p5:ca120108a8b18e5deb186db4ff0a0783f13be9ec340a609529c550cd565b3900	
drone_sd:images	GPS within flight bounding box (lat 39.9574, lon 106.2177)
p5:156179a4c036f4ff09313fa145042228cdcee0792ea1687dd2bfb714a472a7ea	
drone_sd:images	GPS within flight bounding box (lat 39.9574, lon 106.2177)
p5:b6ccdc3903027f542d8fd0379c4f1001f994c638a5d5030195bbc8a7a14468fc	
drone_sd:images	GPS within flight bounding box (lat 39.9587, lon 106.2203)
p5:ba6f37f102e847211489b5371fdb817329d8555f566a132e310763588a20b20e	
drone_sd:images	GPS within flight bounding box (lat 39.9607, lon 106.2209)
p5:01d51be119db0b02d8d7b7e31a01b115411c26dfe177eb66b5dfc3de6d60868c	
drone_sd:images	GPS within flight bounding box (lat 39.9607, lon 106.2209)
p5:aa97f908554705972fbb0362d863056286f863d4c5b17bba34190f4440f90f8b	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:079838ccc4ed59bd93bc6aaa74a337a092063749764e9ee9326df3b3f769107d	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:e00139d2a43b8ae05712f06bc14e6a0f82e64869b9c182f61c61afb888109422	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)

p5:0182321ece95430b6c9dbf7e27a49956dc9761c24f5cc9b03be0a9c47ad25c6d	
drone_sd:images	GPS within flight bounding box (lat 39.9638, lon 106.2208)
p5:74c0691b437ec9e3911cbdf5de1b6654c4f80a36e540b128dbb02f993df566e	
drone_sd:images	GPS within flight bounding box (lat 39.9638, lon 106.2208)
p5:f3ebdeef41ffd3b5ccf62ec7468f6f9b7d86a541bda848ecb4dd9c162f4e49e5	
drone_sd:images	GPS within flight bounding box (lat 39.9638, lon 106.2208)
p5:54922dafa9933d3fe0fb2ae7f657b3fb1d41c9516334e19e6a1b486fd0190993	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:fbc22eb8455bd1f350c4a9fc9df97770072fcc727a354d2434dc1550b0b5b3bd	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:8d90018ed19e66acde8b90016826607c5416c0e38414b5e567e6b19d7815aa57	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:9041aae2121b0393c0d4c0ddf3e2f15b4f0df5249b4179431dd22c46046b416b	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:4169ef410cc02adc6c5eb68509a75b3da17608efc19b0ab98e77813c9cd648a0	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:beb7b3173644b5a8245f1ae715eda90da3639da617a4922b5afd35416f36fef8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd5bdf02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:583321875f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 663b1abdf3b629314072fdea1bc2fe59525f7ca84fa8831297f90ca3bcebbab
Drone Log Name: FLY003.csv
Start: 2017-06-28T17:55:22.762000+00:00
End: 2017-06-28T17:59:12.322000+00:00
Duration: 229.6 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

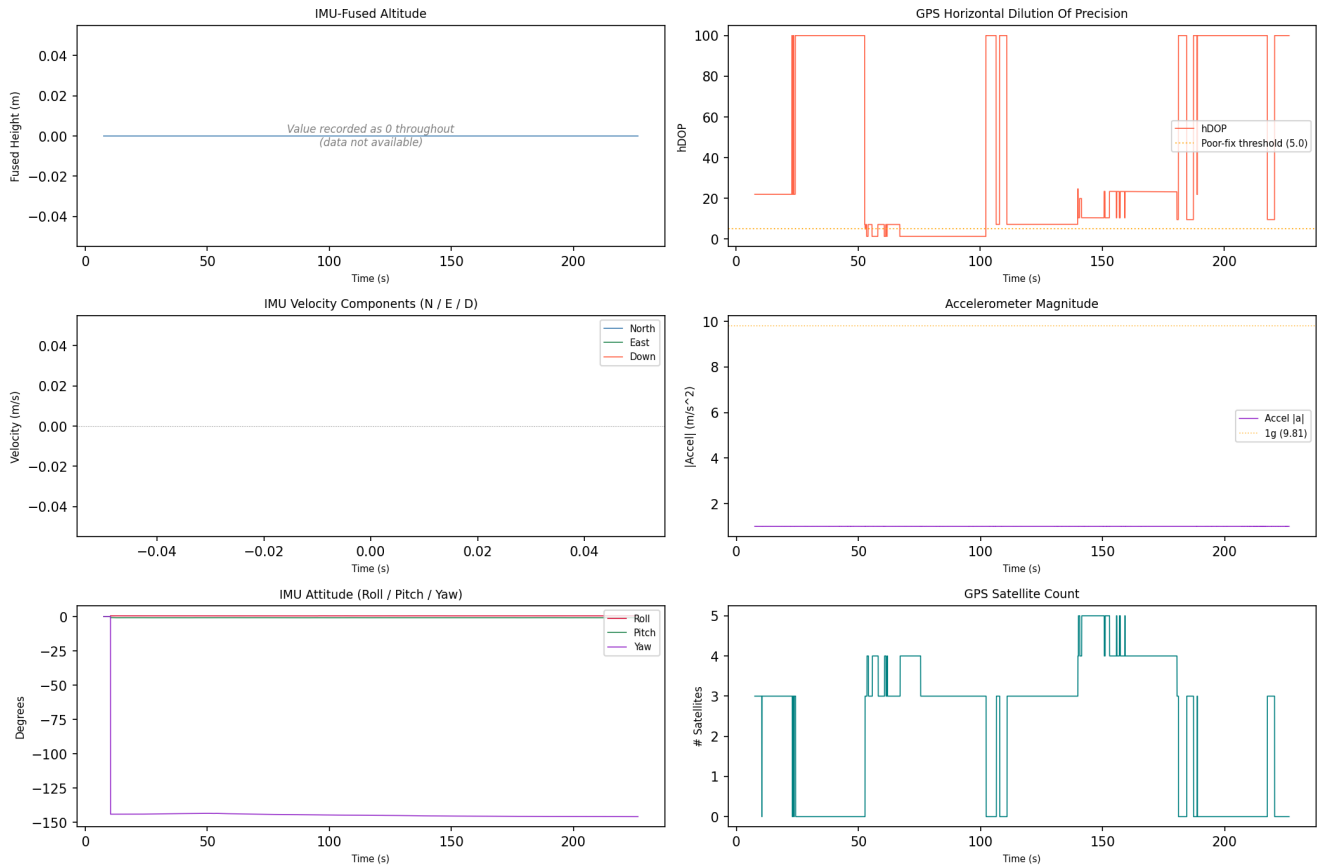
1. Atti
2. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-06-28 17:56:29	drone_flight_storage:dl	Log started	P4P
2017-06-28 17:59:12	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 5a17c8e3c017d7e974ddaabc19be4e5fb6b30c45d240c0ea3ab587393ed8df5b
Drone Log Name: FLY005.csv
Start: 2017-06-29T17:01:02.991000+00:00
End: 2017-06-29T17:06:44.439000+00:00
Duration: 341.4 s

Flight Metrics

Peak Height: 0.303917021407829 m
Distance Travelled: 6.8991329738849965 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 17:01:40	drone_flight_storage:dl	Log started	P4P
2017-06-29 17:02:25	drone_flight_storage:dl	Motor turned on	on
2017-06-29 17:02:30	drone_flight_storage:dl	Motor turned off	off
2017-06-29 17:02:33	drone_flight_storage:dl	Motor turned on	on
2017-06-29 17:02:36	drone_flight_storage:dl	Motor turned off	off
2017-06-29 17:03:21	drone_flight_storage:dl	Reached peak height	0.3 m
2017-06-29 17:06:45	drone_flight_storage:dl	Log ended	

Correlated Artefacts

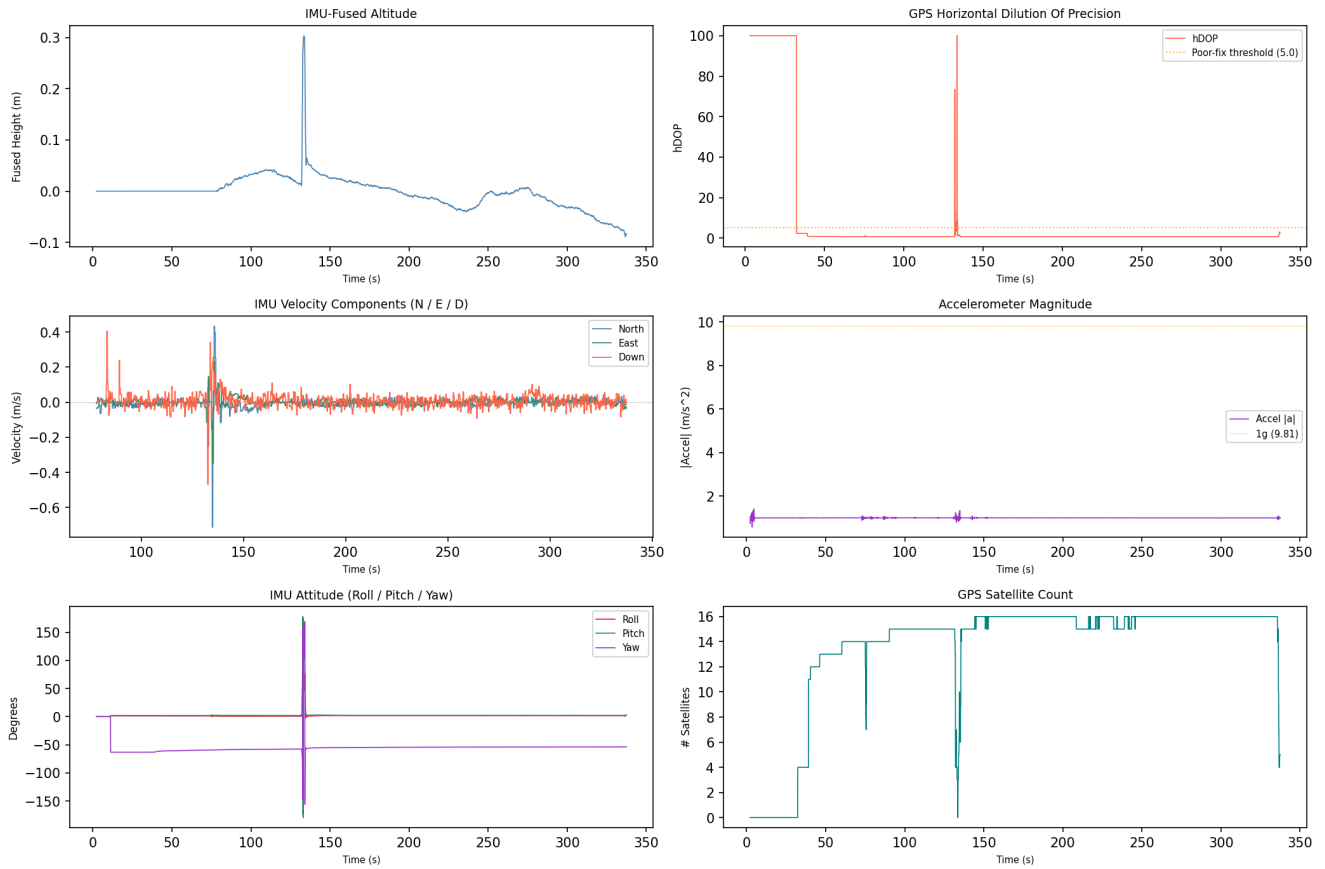
Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9574, lon -106.2177)
p5:9da5fd3a57eba8acf6c7c94462b7edb23eeb78287511457c684373c955f935ba	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd9b9d02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:5833218757f0835eb44faf533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	

drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Drone Log - Sensor Telemetry

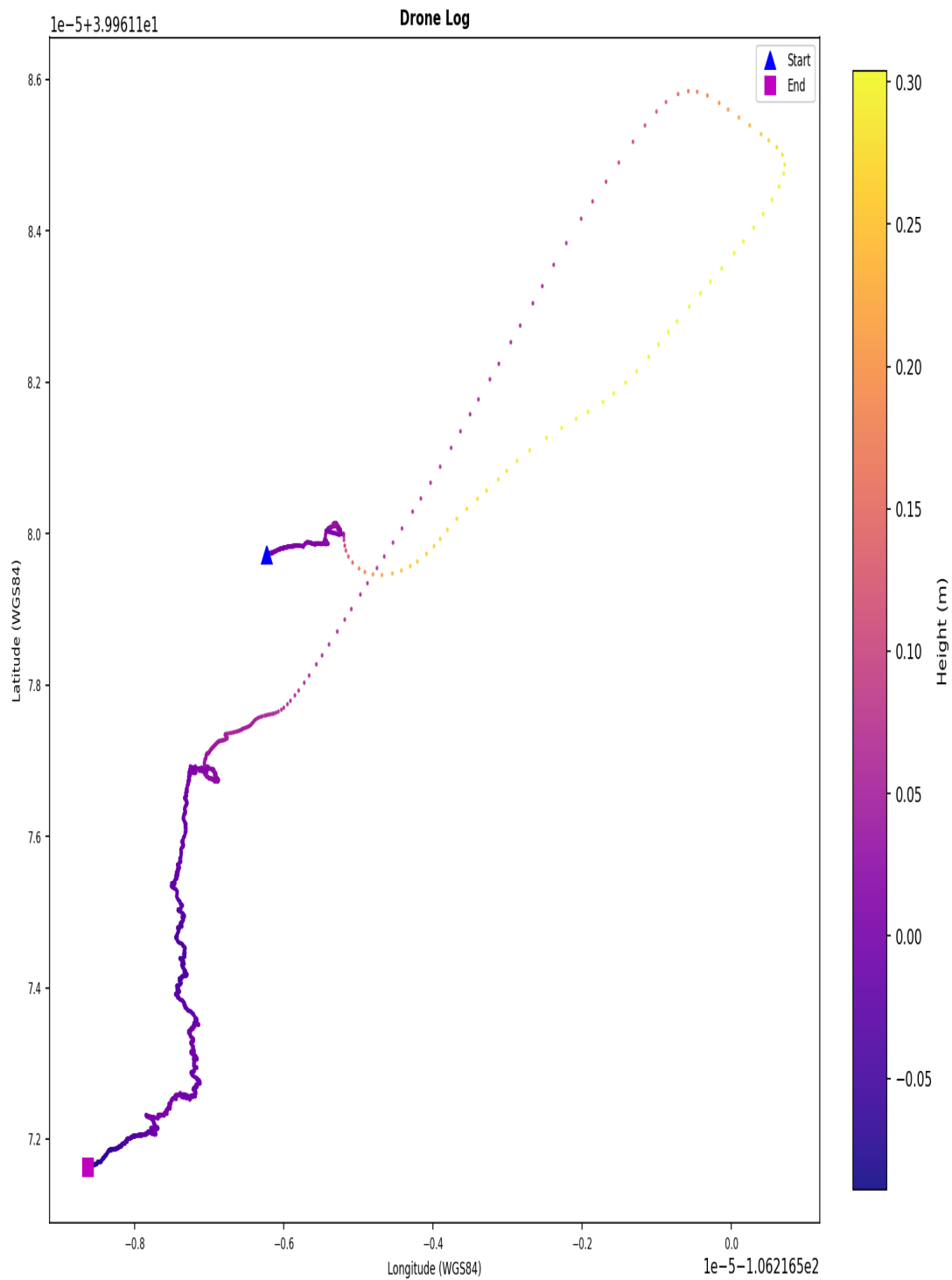
Drone Log Telemetry - flight_03



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_03



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY005.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY005.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 9183bc9d4f904893f1a386ab8c7275475615f80a46cf888d66a542ea8c3276fd
Drone Log Name: FLY006.csv
Start: 2017-06-29T18:24:47.273000+00:00
End: 2017-06-29T18:30:07.551000+00:00
Duration: 320.3 s

Flight Metrics

Peak Height: 1.0282869913420136 m
Distance Travelled: 9.368021331266068 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

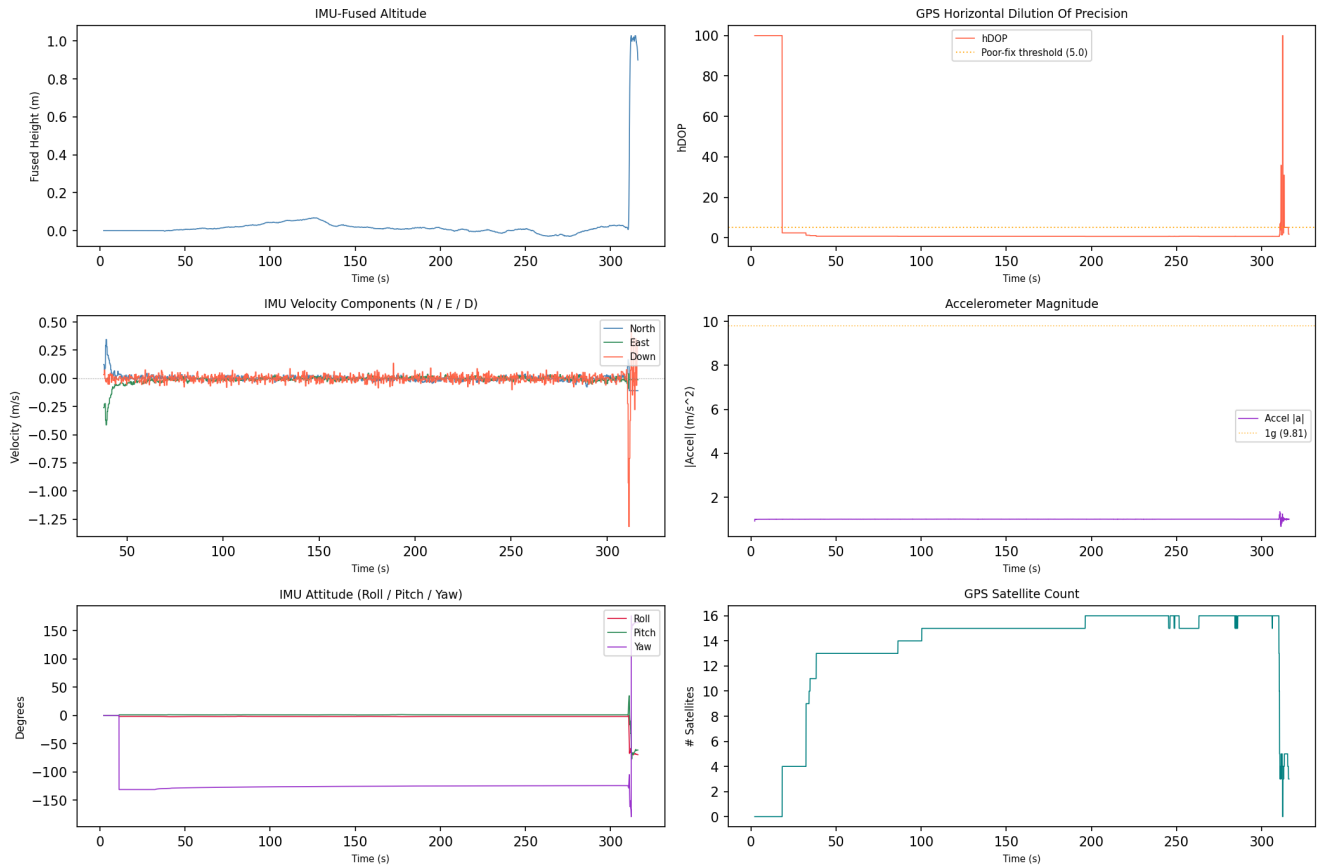
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:25:10	drone_flight_storage:dl	Log started	P4P
2017-06-29 18:30:03	drone_flight_storage:dl	Reached peak height	1.0 m
2017-06-29 18:30:07	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

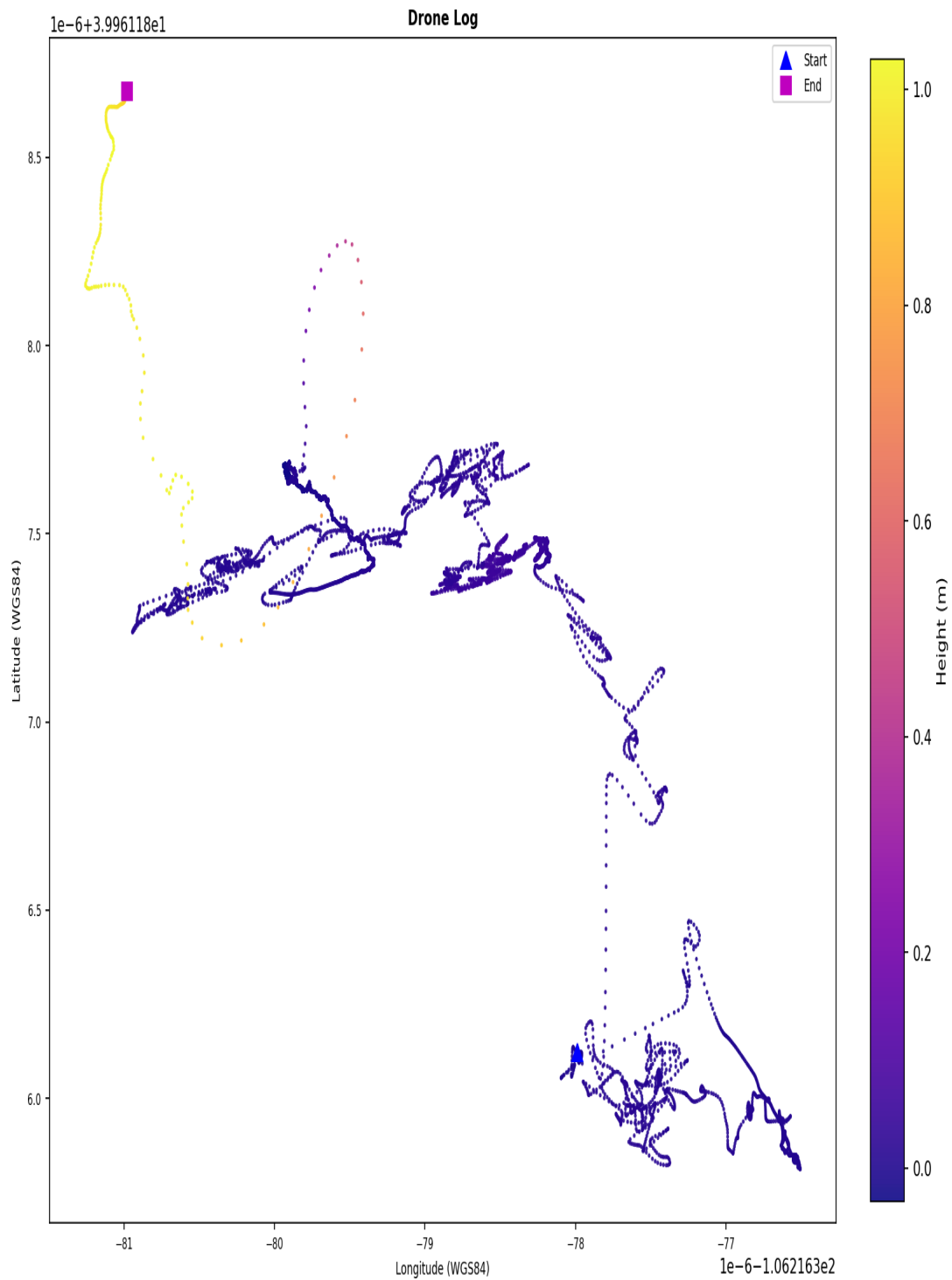
Drone Log Telemetry - flight_04



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_04



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY006.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	high
GPS Overlap:	725.2 s
Median GPS Distance:	4.57 m
Overlap:	94%
GPS Match Rate:	93%
Flight Record SHA-256:	869db0c19c3e5e6bbe694ad2159ed2b617d4d80ba53eb8725fe78f3cc8ef6ec3
Flight Record Name:	DJIFlightRecord_2017-06-29_[12-39-25].csv
Start:	2017-06-29T18:39:25.651000+00:00
End:	2017-06-29T18:52:14.092000+00:00
Duration:	768.4 s
Drone Log SHA-256:	77160f1321cc9fdf29ff41d000d57d1b6af3d05b52927da27edff23c9c53b74d
Drone Log Name:	FLY007.csv
Start:	2017-06-29T18:38:41.869000+00:00
End:	2017-06-29T18:51:30.881000+00:00
Duration:	769.0 s

Flight Metrics

Peak Height:	120.57006692179861 m
Distance Travelled:	1715.926248983839 m
Video Recording:	No
Photo Capture:	No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. EngineStart
7. AssitedTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:39:10	drone_flight_storage:dl	Log started	P4P
2017-06-29 18:39:23	drone_flight_storage:dl	Motor turned on	on
2017-06-29 18:39:25	ctrl_and:fr (v4.1.2)	Log started	SN: 07JDE580020139; App: 4.1.2; P4 Professional
2017-06-29 18:39:25	ctrl_and:fr (v4.1.2)	Motor turned on	on
2017-06-29 18:48:44	drone_flight_storage:dl	Reached peak height	120.6 m
2017-06-29 18:50:12	ctrl_and:fr (v4.1.2)	Reached peak height	120.2 m
2017-06-29 18:51:21	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:51:21	drone_flight_storage:dl	Motor turned on	on
2017-06-29 18:51:23	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:51:31	drone_flight_storage:dl	Log ended	
2017-06-29 18:52:14	ctrl_and:fr (v4.1.2)	Log ended	Dist home: 2.0 m; Height: -1.8 m; Rec: 573.0 s

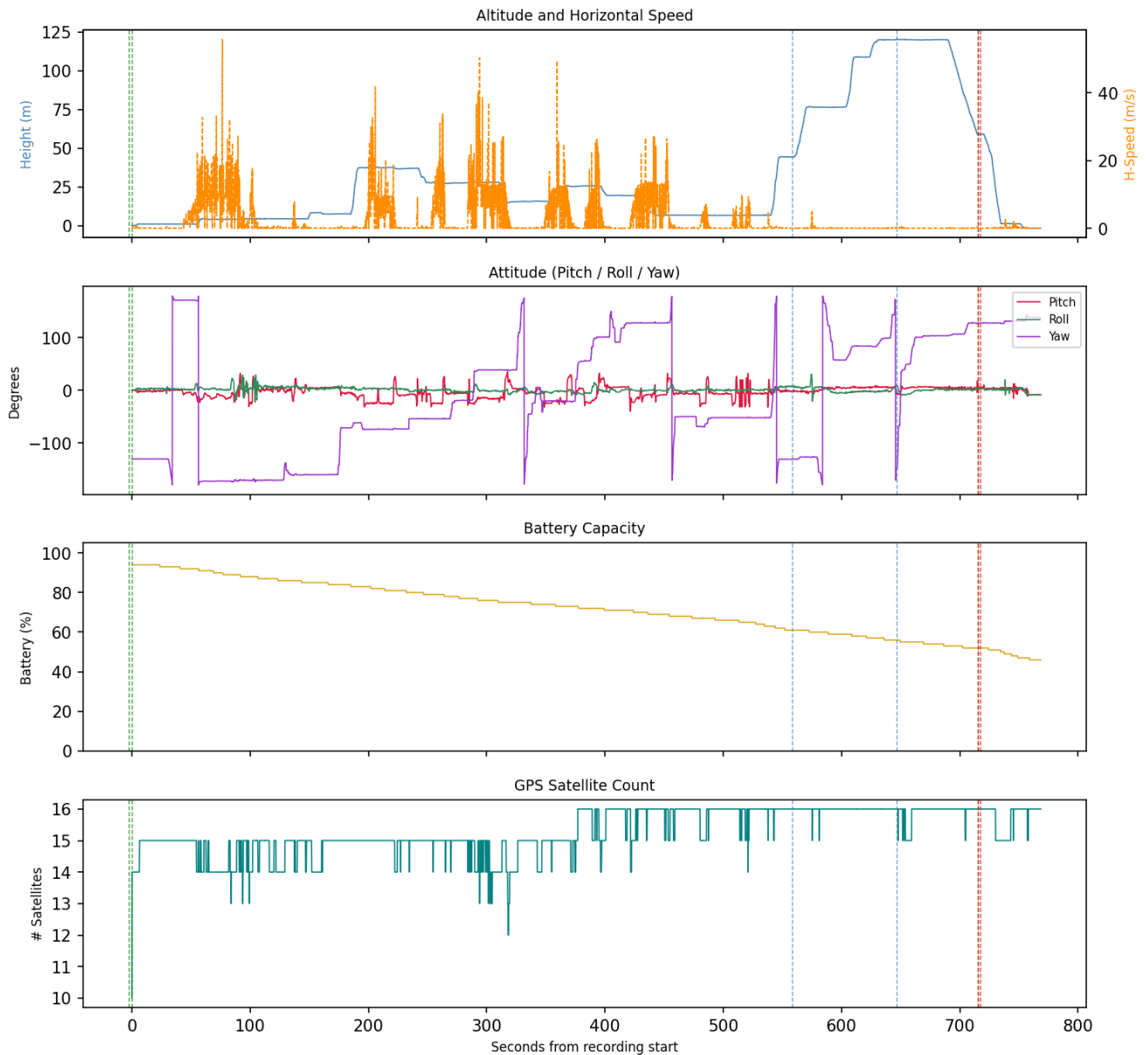
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9574, lon -106.2177)
p5:9da5fd3a57eba8acf6c7c94462b7edb23eeb78287511457c684373c955f935ba	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbdbbfd02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:583321875f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Flight Record - Sensor Telemetry

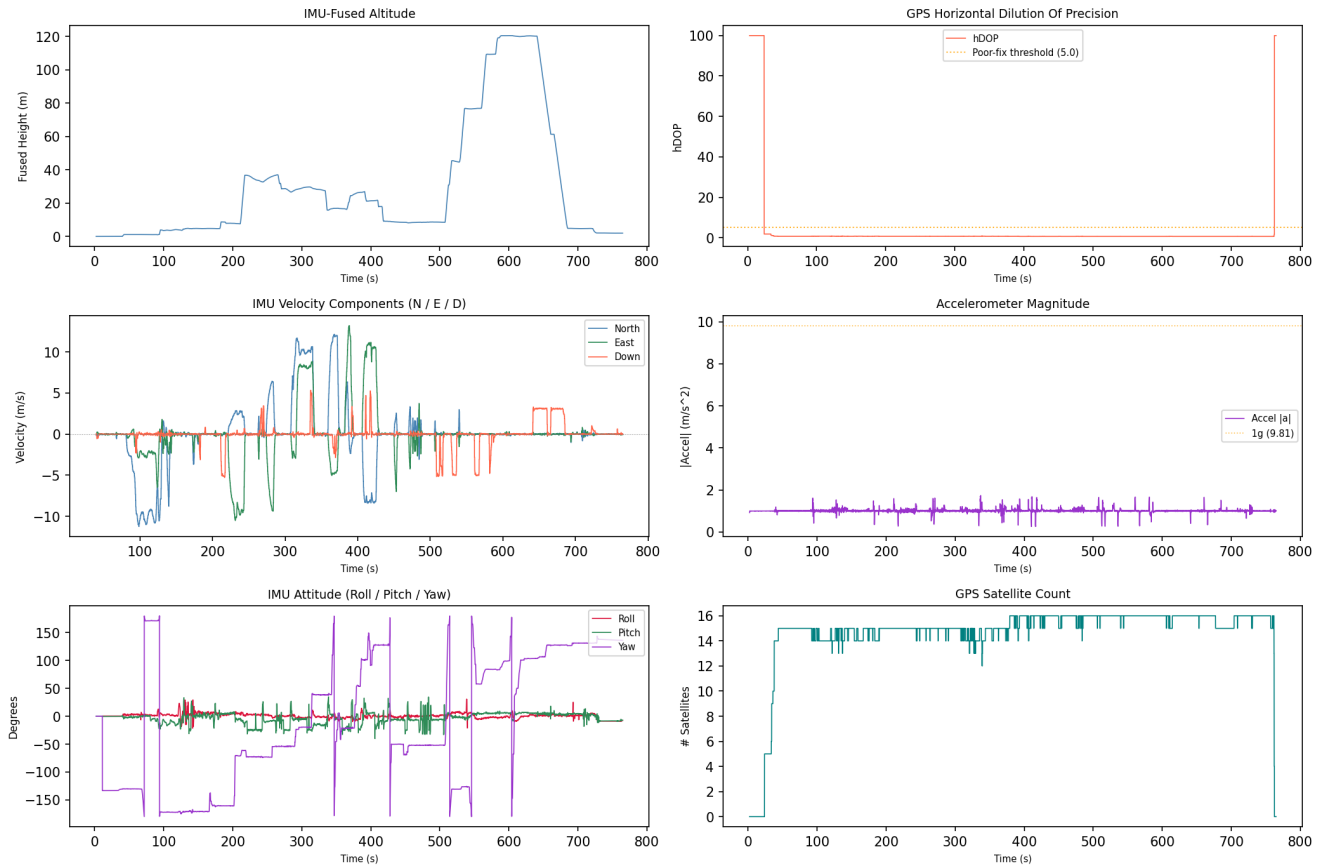
Flight Record Telemetry - flight_05



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

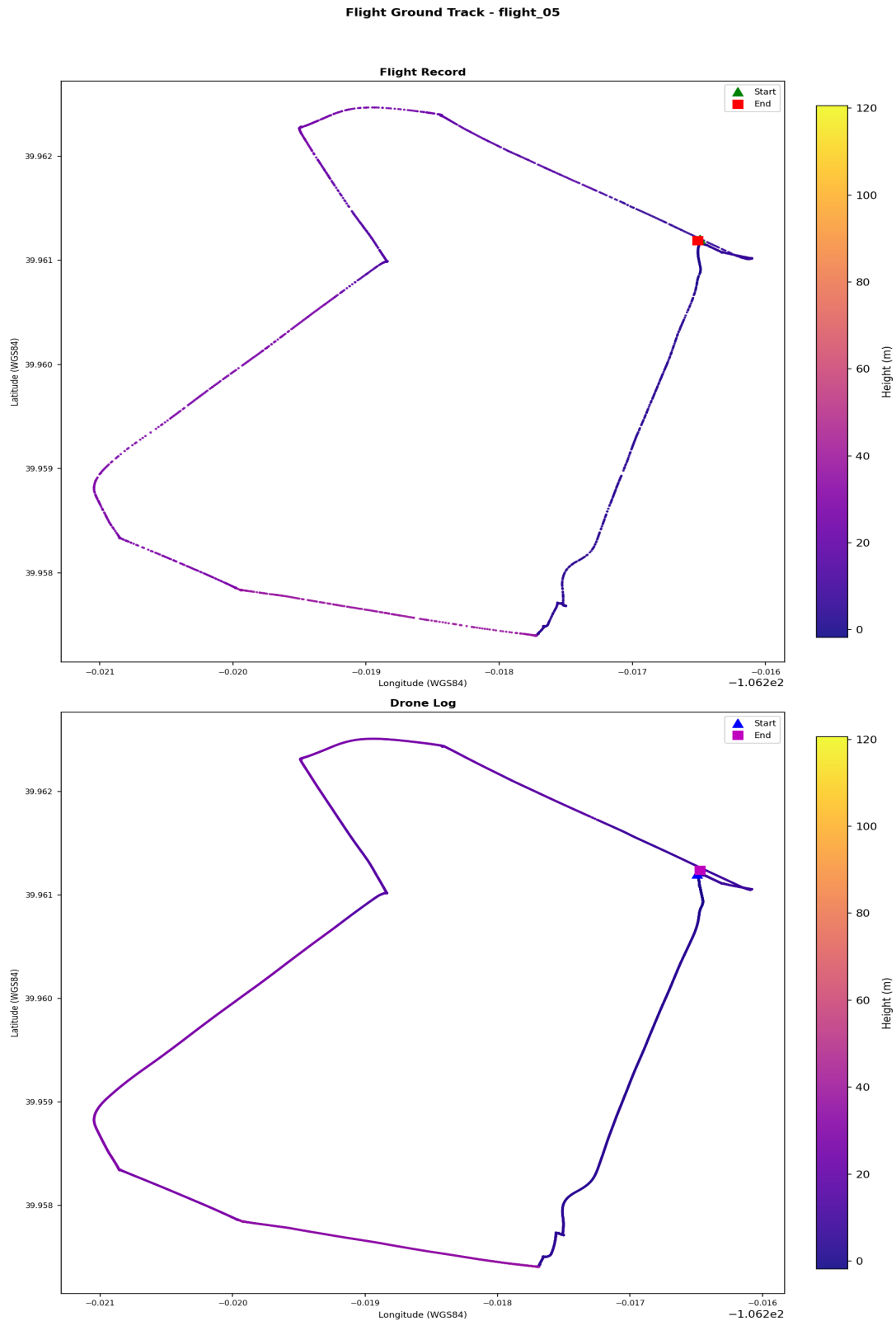
Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_05



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY007.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 3d86188a08d9457e427049720d25658b8ab5695c4ac6037b6cf86b4824f28379
Flight Record Name: DJIFlightRecord_2017-06-29_[12-52-14].csv
Start: 2017-06-29T18:52:14.647000+00:00
End: 2017-06-29T18:52:15.571000+00:00
Duration: 0.9 s

Flight Metrics

Peak Height: -
Distance Travelled: 0.03 m
Video Recording: No
Photo Capture: No
Photos Taken: 0
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

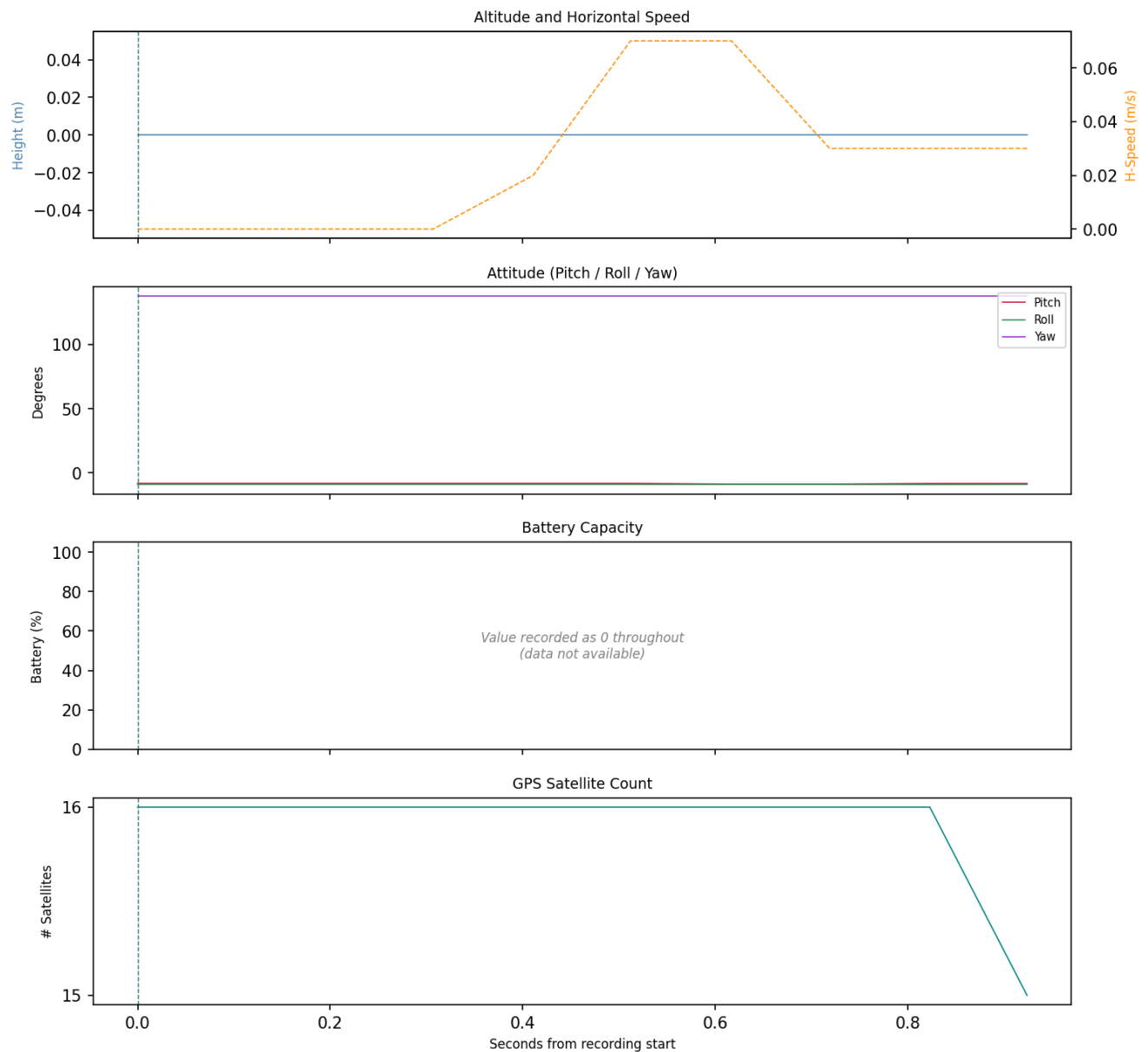
1. EngineStart
2. AssistedTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-29 18:52:14	ctrl_and:fr (v4.1.2)	Log started	App: 4.1.2; P4 Professional
2017-06-29 18:52:14	ctrl_and:fr (v4.1.2)	Reached peak height	0.0 m
2017-06-29 18:52:14	ctrl_and:fr (v4.1.2)	Motor turned on	on
2017-06-29 18:52:15	ctrl_and:fr (v4.1.2)	Log ended	Dist home: 0.0 m; Height: 0.0 m; Rec: 0.0 s

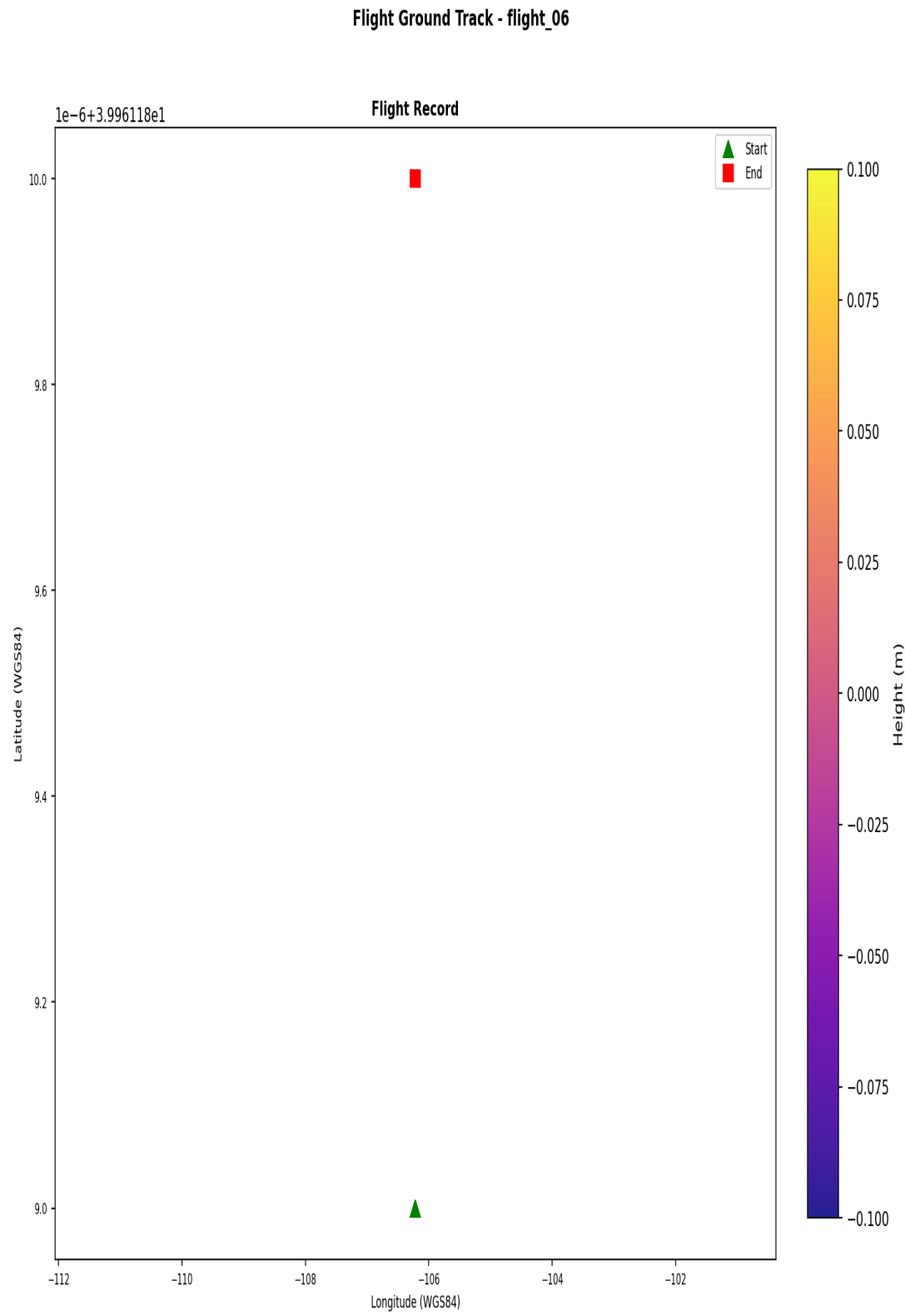
Flight Record - Sensor Telemetry

Flight Record Telemetry - flight_06



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 69ddbdf905e27dbf84c47afcaff5eec7cf5cb3f4ccd58951d8e2206874310721
Drone Log Name: FLY008.csv
Start: 2017-06-29T20:26:17.019000+00:00
End: 2017-06-29T20:49:56.669000+00:00
Duration: 1419.7 s

Flight Metrics

Peak Height: 115.6851035468051 m
Distance Travelled: 2264.5885478447535 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 20:26:55	drone_flight_storage:dl	Log started	P4P
2017-06-29 20:35:16	drone_flight_storage:dl	Motor turned on	on
2017-06-29 20:46:56	drone_flight_storage:dl	Reached peak height	115.7 m
2017-06-29 20:49:12	drone_flight_storage:dl	Motor turned off	off
2017-06-29 20:49:57	drone_flight_storage:dl	Log ended	

Correlated Artefacts

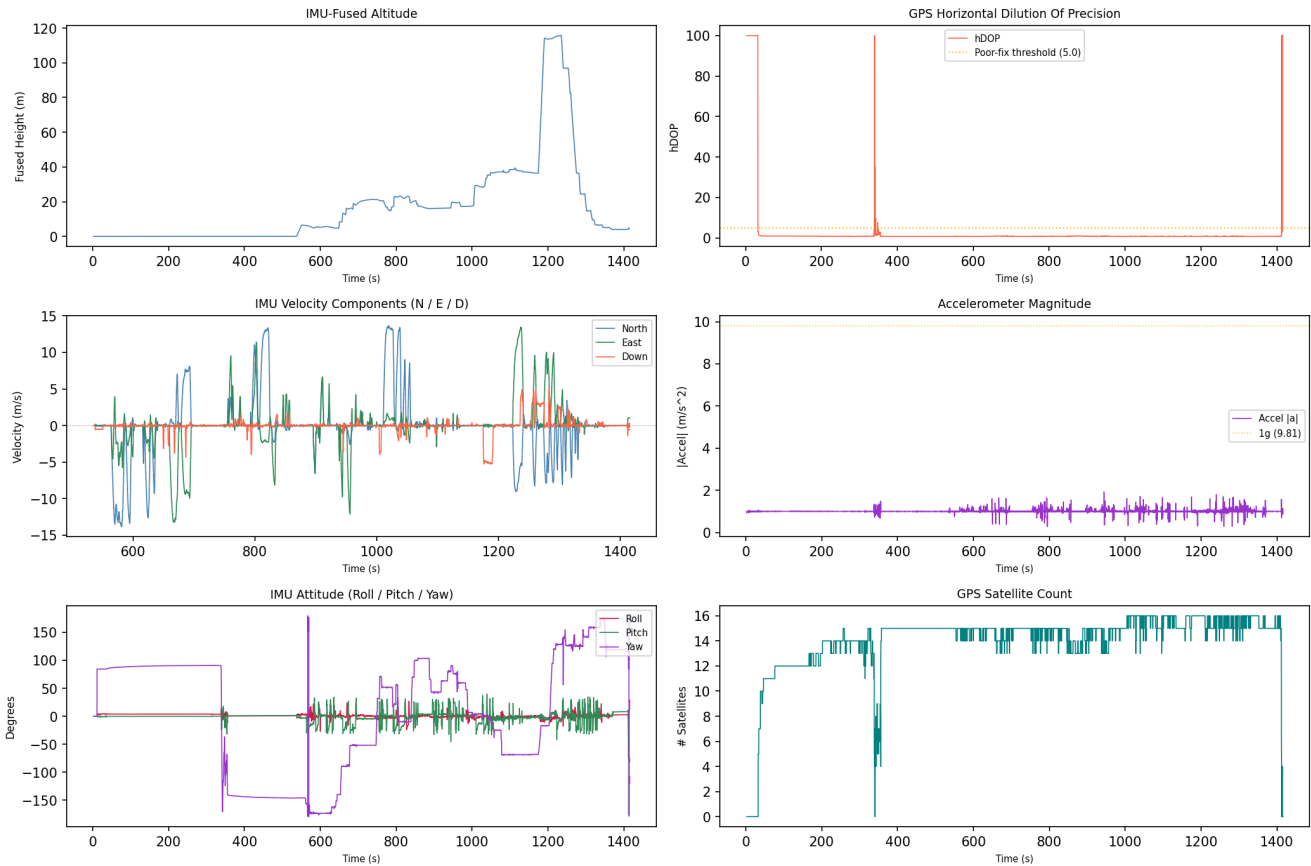
Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9574, lon -106.2177)
p5:9da5fd3a57eba8acf6c7c94462b7edb23eeb78287511457c684373c955f935ba	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbdbbfd02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:5833218757f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514	

drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Drone Log - Sensor Telemetry

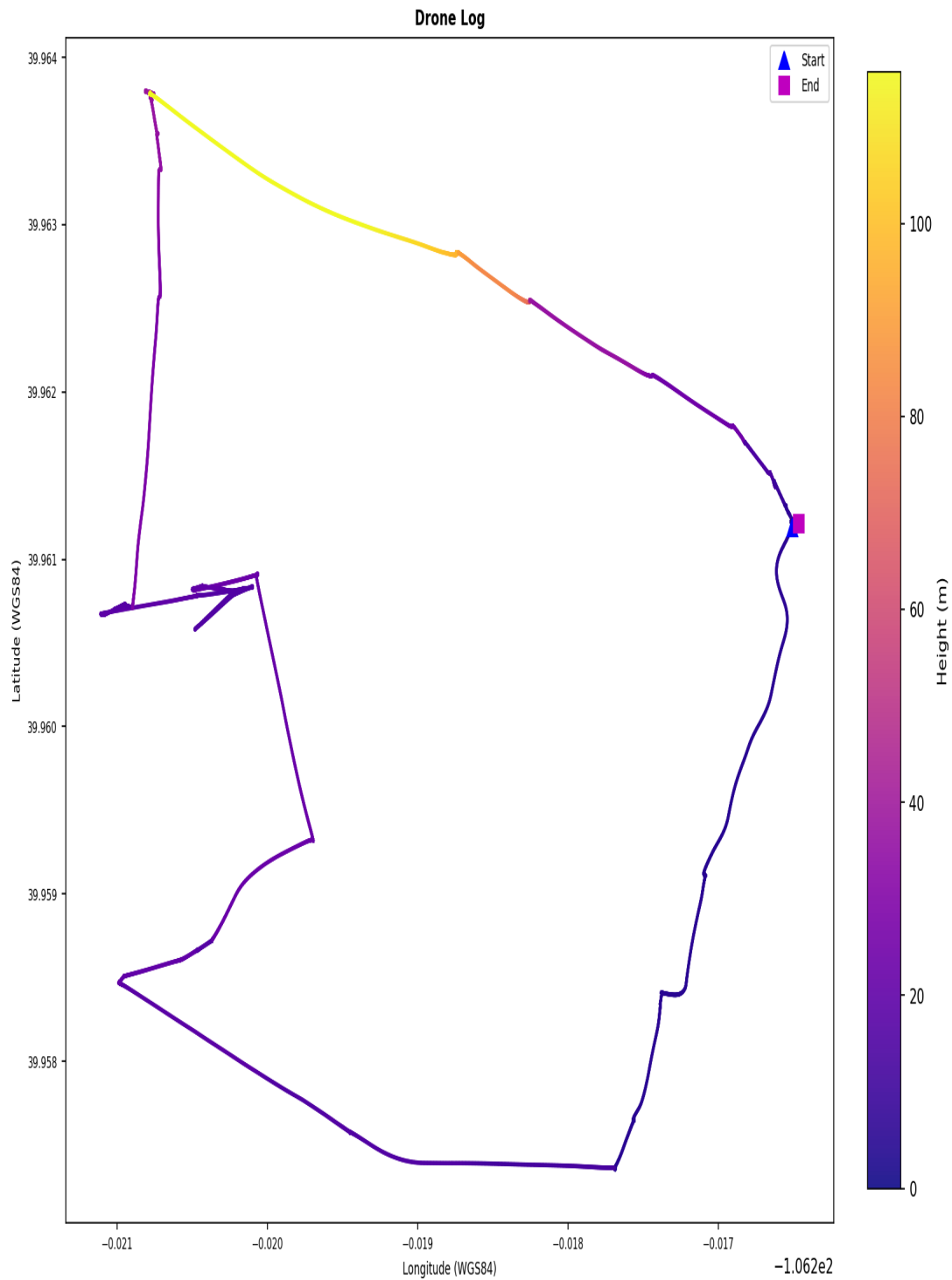
Drone Log Telemetry - flight_07



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_07



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY008.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	2	7	0	40	2	2	2
drone_sd	0	0	0	0	0	17	12
drone_flight_storage	0	0	11	0	0	0	0

Data Quality Notes per Category

Drone Logs (11 files):

- 11 file(s) with row anomalies
- 11 file(s) with empty columns
- 11 file(s) with constant-value columns

Flight Logs (40 files):

- formats: 2 bracketed_log, 1 heading_log, 37 unknown

Flight Records (2 files):

- 1 file(s) with row anomalies
- 2 file(s) with empty columns
- 2 file(s) with constant-value columns

Images (19 files):

- 17 file(s) contain(s) EXIF GPS coordinates
- 19 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

Videos (14 files):

- 13 file(s) contain(s) EXIF GPS coordinates
- 13 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 11 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 11 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 11 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: log files present in the following format(s): 2 bracketed_log, 1 heading_log, 37 unknown.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 2 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 2 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for DF006_flight.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for DF006_flight.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller ios]: source evidence not found
7. [p4 - controller ios]: source evidence not found
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY010.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Drone Logs

norm_FLY000.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 5 row(s) affected
- zero_coordinate: 272 row(s) affected
- gps_poor_fix: 272 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 133 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+128 more)
- gps_accuracy_poor: 272 row(s) affected

norm_FLY001.csv

- missing_gps: 15 row(s) affected
- zero_coordinate: 1845 row(s) affected
- gps_poor_fix: 14467 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 98 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+93 more)
- gps_accuracy_poor: 16078 row(s) affected
- quaternion_invalid: 244 row(s) affected

norm_FLY002.csv

- missing_gps: 7 row(s) affected
- zero_coordinate: 1984 row(s) affected
- gps_poor_fix: 16196 row(s) affected
- contains_no_value: 18 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+13 more)
- contains_constant_value: 91 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D,

AirSpeed:air_vby:D (+86 more)

- gps_accuracy_poor: 17185 row(s) affected
- quaternion_invalid: 253 row(s) affected

norm_FLY003.csv

- missing_gps: 8 row(s) affected
- gps_poor_fix: 4726 row(s) affected
- contains_no_value: 18 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+13 more)
- contains_constant_value: 126 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+121 more)
- gps_accuracy_poor: 5116 row(s) affected
- quaternion_invalid: 79 row(s) affected

norm_FLY004.csv

- missing_gps: 16 row(s) affected
- zero_coordinate: 3635 row(s) affected
- gps_poor_fix: 6595 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 108 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+103 more)
- gps_accuracy_poor: 6595 row(s) affected
- quaternion_invalid: 234 row(s) affected

norm_FLY005.csv

- missing_gps: 18 row(s) affected
- zero_coordinate: 781 row(s) affected
- gps_poor_fix: 861 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, osd_data:lowVoltage
- contains_constant_value: 74 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+69 more)
- gps_accuracy_poor: 874 row(s) affected
- quaternion_invalid: 248 row(s) affected
- attitude_out_of_bounds: 41 row(s) affected

norm_FLY006.csv

- missing_gps: 7 row(s) affected
- zero_coordinate: 451 row(s) affected
- motor_airborne_off: 82 row(s) affected
- gps_poor_fix: 485 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, osd_data:lowVoltage
- contains_constant_value: 88 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+83 more)
- gps_accuracy_poor: 555 row(s) affected
- quaternion_invalid: 249 row(s) affected

norm_FLY007.csv

- missing_gps: 15 row(s) affected
- zero_coordinate: 597 row(s) affected
- motor_airborne_off: 229 row(s) affected
- gps_poor_fix: 669 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost (+1 more)
- contains_constant_value: 55 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl_f:D (+50 more)
- gps_accuracy_poor: 669 row(s) affected
- quaternion_invalid: 244 row(s) affected

norm_FLY008.csv

- missing_gps: 14 row(s) affected
- zero_coordinate: 432 row(s) affected
- motor_airborne_off: 1279 row(s) affected
- gps_poor_fix: 947 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 55 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+50 more)
- gps_accuracy_poor: 998 row(s) affected
- quaternion_invalid: 240 row(s) affected
- attitude_out_of_bounds: 26 row(s) affected

norm_FLY009.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 290 row(s) affected
- gps_poor_fix: 290 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 118 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+113 more)
- gps_accuracy_poor: 290 row(s) affected
- quaternion_invalid: 246 row(s) affected

norm_FLY010.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 10853 row(s) affected
- gps_poor_fix: 10853 row(s) affected
- contains_no_value: 24 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+19 more)
- contains_constant_value: 103 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+98 more)
- gps_accuracy_poor: 10853 row(s) affected
- quaternion_invalid: 251 row(s) affected
- attitude_out_of_bounds: 22 row(s) affected

Flight Logs**29-06-2017**

- format: heading_log | entries: 16

log-2017-06-28.txt

- format: unknown | entries: 0

log-2017-06-28_1.txt

- format: unknown | entries: 0

log-2017-06-28_10.txt

- format: unknown | entries: 0

log-2017-06-28_11.txt

- format: unknown | entries: 0

log-2017-06-28_12.txt

- format: unknown | entries: 0

log-2017-06-28_13.txt

- format: unknown | entries: 0

log-2017-06-28_14.txt

- format: unknown | entries: 0

log-2017-06-28_15.txt

- format: unknown | entries: 0

log-2017-06-28_16.txt

- format: unknown | entries: 0

log-2017-06-28_17.txt

- format: unknown | entries: 0

log-2017-06-28_18.txt
- format: unknown | entries: 0

log-2017-06-28_2.txt
- format: unknown | entries: 0

log-2017-06-28_3.txt
- format: unknown | entries: 0

log-2017-06-28_4.txt
- format: unknown | entries: 0

log-2017-06-28_5.txt
- format: unknown | entries: 0

log-2017-06-28_6.txt
- format: unknown | entries: 0

log-2017-06-28_7.txt
- format: unknown | entries: 0

log-2017-06-28_8.txt
- format: unknown | entries: 0

log-2017-06-28_9.txt
- format: bracketed_log | entries: 6

log-2017-06-29.txt
- format: unknown | entries: 0

log-2017-06-29_1.txt
- format: unknown | entries: 0

log-2017-06-29_10.txt
- format: unknown | entries: 0

log-2017-06-29_11.txt
- format: bracketed_log | entries: 2

log-2017-06-29_12.txt
- format: unknown | entries: 0

log-2017-06-29_13.txt
- format: unknown | entries: 0

log-2017-06-29_14.txt
- format: unknown | entries: 0

log-2017-06-29_15.txt
- format: unknown | entries: 0

log-2017-06-29_16.txt
- format: unknown | entries: 0

log-2017-06-29_17.txt
- format: unknown | entries: 0

log-2017-06-29_18.txt
- format: unknown | entries: 0

log-2017-06-29_19.txt
- format: unknown | entries: 0

log-2017-06-29_2.txt
- format: unknown | entries: 0

log-2017-06-29_3.txt
- format: unknown | entries: 0

log-2017-06-29_4.txt
- format: unknown | entries: 0

log-2017-06-29_5.txt
- format: unknown | entries: 0

log-2017-06-29_6.txt

- format: unknown | entries: 0

log-2017-06-29_7.txt

- format: unknown | entries: 0

log-2017-06-29_8.txt

- format: unknown | entries: 0

log-2017-06-29_9.txt

- format: unknown | entries: 0

Flight Records

norm_DJIFlightRecord_2017-06-29_[12-39-25].csv

- altitude_negative: 147 row(s) affected
- altitude_spike: 1 row(s) affected
- contains_no_value: 54 column(s) - APP_GPS.latitude, APP_GPS.longitude, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto (+49 more)
- contains_constant_value: 150 column(s) - APP_GPS.accuracy, APP_SER_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.errorType, CENTER_BATTERY.fullCapacity [mAh] (+145 more)

norm_DJIFlightRecord_2017-06-29_[12-52-14].csv

- contains_no_value: 150 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, APP_WARN.warn (+145 more)
- contains_constant_value: 106 column(s) - APP_TIP.tip, CALC.distance [m], CALC.distance.running_max [m], CALC.height.running_max [m], CUSTOM.hSpeed [m/s] (+101 more)

Images

thumb_49cbe7a3baed60c7_1498761563000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_7d5a8622435fafbd_1498761669000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0007.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0008.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0010.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0013.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0014.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0015.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0016.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0017.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0018.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0019.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0020.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0021.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0022.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0023.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

Videos

2017_06_29_12_41_46.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0002.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Drone Model inconsistency detected - conflicting values observed: P4 Professional, P4P. Manual verification required.
4. Note: no controller device found - Device and account information.
5. Note: no firmware version found - Device and account information.
6. Note: no aircraft model code found - Device and account information.
7. Note: no device uuid found - Device and account information.
8. 6 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_06, flight_07). Independent corroboration is absent for these flight(s).
9. 7 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.go.v4.xml	Account Data	controller_android	extract_logical
SHA-256:	5c93def876a07079d25a3788127cd503013cf5f3cd510f61f17a4118916850a8		
SHA-1:	f0147aa68b6ca0847ecee13e21f383f3f364ac0e		

dji.pilot.xml	Account Data	controller_android	extract_logical
SHA-256:	d306c329fc2d1b3fb753806317f5f81bee156160bf9dc6acb50d18e5e4961c79		
SHA-1:	8cd558d7b69f1b8994dd1c0af58399df41557f54		

dji.db	Databases	controller_android	extract_logical
SHA-256:	575f5edb0edafe24c7350eccf497cb9eafa2915a7a8f56d3ea2af845470ab8c6		
SHA-1:	ffc21882d1a34f1662bda432f71d111f6679d1bd		

dji_1.db	Databases	controller_android	extract_logical
SHA-256:	03893b82908e748ec88dd1fe47fe5c8ce39e3a63bea63401eda62760e2d669c1		
SHA-1:	8a131b3c180b11fd0451afae0c9ff39ae8130a17		

flysafe_polygon_1860.db	Databases	controller_android	extract_logical
SHA-256:	de615b0a8e225bc1a815b94308e7d1644b648995c43bc0af075f0990adadb90d		
SHA-1:	220f2c8924ad0b196f233dc26b4adfb30367d539		

jpush_local_notification.db	Databases	controller_android	extract_logical
SHA-256:	c785ff31316c80b481d1ba6da796f401253cd17c812b3a41988c8aa9faea89b6		
SHA-1:	3d1cb1b9a2f2d533e0a84d01c7a4e3f9c115c85d		

jpush_statistics.db	Databases	controller_android	extract_logical
SHA-256:	b8ed4d31153bcc6fdcaa7004f545aa244155e0d521db130f3c29c522e7d39ad		
SHA-1:	263b449bf8a8b9f2eb3b44aa55b94798862408e9		

logdb.db	Databases	controller_android	extract_logical
SHA-256:	90b607612fb1bc79f1d604727803ddf6161d5d56ee3afbe04989a66183d938a9		
SHA-1:	32b57c3782cfdaaea4f8cba76c599ae8dcd04727		

service.db	Databases	controller_android	extract_logical
SHA-256:	2f839d8db239fbac29b0c5e4e12561a35331368506f12a70478ba75a9c16a5b5		
SHA-1:	f1db295bb6b5ba9b3522d63eb4469ccba6641acb		

FLY000.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	40910fa3e293ceff72384866cefc2d8cfe445b369108eb650ee253fa34756d62		
SHA-1:	a1a6b5437d8be2c57fb96d9f17c4409b719dea5a		

FLY001.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	3b9e1db78a6933c0b6647abe6b1367de45226050a6d13a0ae237772746b80dd3		
SHA-1:	66b20d6cdf2aaffd82e8e8a27a6f89f4e014d67b		

FLY002.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	078e6f73bb39451672e0550eb1f0b266abd2eabed6ba13bb38ffd58d73194afe		
SHA-1:	b4be20bee5159cffaf27762f0c43ec591f340f53		

FLY003.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8f67c39f42b98b3e27ac5b72eef129e59a9db0602aa0a95a99f3579ca7f944ca			
SHA-1:	16f0b9b93e1d3ebae90e914743347ef2bc722523			

FLY004.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	2202fdaf6407238f1bb1a3656dc1e41244c6a06ce59242f877564015ef82e360			
SHA-1:	6899942028c02d83021284560f4900179a84fdcc			

FLY005.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c139fdc624b47bd2d6caa8d9626aeaa3bcf64e7e43ded05bddcf5724a9c4785b			
SHA-1:	c1514bc1487795d7f36c30ca41191b44ced4ca2d			

FLY006.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8efcb686b413ecac6d39b10502c299ffb32a0173263eaae6a918faa04eb1c757			
SHA-1:	d6967f4fc5e9ef15e62b1c652ced2859876e3159			

FLY007.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	34efb6cb1096929488198b989e69f93447f57862ba6a2454c058e11e04c8f469			
SHA-1:	cf61cf4e317590c21ec1d6ca57c45514d7eae4ec			

FLY008.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	f5b639c7336f87d170212acbe4fa43aeca30262f2593ec52e6643e5841cca741			
SHA-1:	b66b76ab3667a709252ed1887b5c2c9722dc3ed1			

FLY009.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	d6708dc80052a52512279b8dc8a9faf4a5c29818d7fcb27a9995a78578642884			
SHA-1:	b5d9bd857406c01630009fdaf140e8202a2430e2			

FLY010.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	6962f2dd63ede85ecad20f31c5d12fc50dbc23c3f4889a969683f961e72990ec			
SHA-1:	251f9e7d60291c3f6d0fd04743e902aad018166f			

29-06-2017		Flight Logs	controller_android	extract_logical
SHA-256:	44f93183c8ead10d279e4bfbd1f63335231cfcf358b222c55cf1a3d7f12679f			
SHA-1:	7cc744bb2b78b14e841ecf5482a93245436c6951			

log-2017-06-28.txt		Flight Logs	controller_android	extract_logical
SHA-256:	3ae17cb9e04a9710aab9861e03525a68cd7da2bd89bc9641e8bf6cad3f8402be			
SHA-1:	f40d3a96c9e9063a955797c77866e4d02127d8ad			

log-2017-06-28_1.txt		Flight Logs	controller_android	extract_logical
SHA-256:	3ba10545ad41d9900a923c36fd9e09305cb8175b3d3610212186fccb26c6e1eb			
SHA-1:	c50e84a7d7905c3eaa78ee28802a05a29cacde23			

log-2017-06-28_10.txt		Flight Logs	controller_android	extract_logical
SHA-256:	e207cd26405075f8c9c0c96ea52b48668fef4e821d597bd3ce3efc4323ae1ec4			
SHA-1:	52fe0d2e3c8af81b4ce68e2bd4aca4c9f32ac97b			

log-2017-06-28_11.txt		Flight Logs	controller_android	extract_logical
SHA-256:	d5070e04937cf5eddc96201f22ef2e34e7d7745ce5b687582bf7edce97459ad1			
SHA-1:	00b2a14a01fc3d8f262f2a236eac1a36f228a2fb			

log-2017-06-28_12.txt	Flight Logs	controller_android	extract_logical
SHA-256:	cc410a7ab4357860afe5f8fcca1022155b0544ef942f21a4a40ecc4163e3274a		
SHA-1:	5d546a5cfa71530a36b1fddf718869e0edae215f		

log-2017-06-28_13.txt	Flight Logs	controller_android	extract_logical
SHA-256:	73572f9c703d00f9cfd5e0eb72f63e6681660de827d79a9284e072528009d5c0		
SHA-1:	8f4cd8227183868a9249345a58f72054ba0e36f5		

log-2017-06-28_14.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9d8c9b0cb45363fecdd819a135ba9928098ad0bc3e10db23e6c067023ab716152		
SHA-1:	f964f0c4324baf3a4d791e873d0855243944407a		

log-2017-06-28_15.txt	Flight Logs	controller_android	extract_logical
SHA-256:	069a9e3d529c567e45992755b51e38072ab181bf8ea3e9787f41c2423bac486a		
SHA-1:	8cc1a1cf0db0c685e30292d4540782ffdcf145a3		

log-2017-06-28_16.txt	Flight Logs	controller_android	extract_logical
SHA-256:	deae27fed6eab97dc62f79ac1a5cd21e995a47e4b5fe88317e4f02f0022b3cbe		
SHA-1:	78eb43fc1562dc03ea976a9ca85e61f5f5e0c69f		

log-2017-06-28_17.txt	Flight Logs	controller_android	extract_logical
SHA-256:	81209455983367d9932fa91d57f97dc1654012ab8578c0a934f862640a8e2951		
SHA-1:	5d18f9a7c09817a9d639ccde0f6aafa7702ed722		

log-2017-06-28_18.txt	Flight Logs	controller_android	extract_logical
SHA-256:	a2fe3aa137f3e749ac743c477abb42d17fb8dc1578427df3d677db7588cb7c2e		
SHA-1:	95cd2dca8cbd3e11eb7e0a24e97d552dc5ea53d9		

log-2017-06-28_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	f7f7483e0face67a7d3e3cb3eacd2968e452053d5ca4c2a474e0da39dfd392b5		
SHA-1:	cd70d20cb01906b947299143ffc530344b340d81		

log-2017-06-28_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	69acb7a10ea414daee695fc43a75f0d6ffd89e20d98f1ec86241520a5af7a125		
SHA-1:	152bcbe88d7c3db5921566b36f90d00165fefc4d		

log-2017-06-28_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	d52326382d6ca2f48b27b99327729f5929b7cd59fb6466ee1f04eca2b57a866a		
SHA-1:	ff76b730c31450482216dbbed3fcf71cba6c4507		

log-2017-06-28_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	f14642c57577ae53b31f37cb540483ee696687e17c25d5de143466fa2caaba09		
SHA-1:	9d7bfd34d794f6479abedec0792045571685b337		

log-2017-06-28_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	782a79125b0cb9f3f35ab039725623f8cd6e5571e0994d729b60475cbb939b20		
SHA-1:	b3a690182bb650c29abd32187801f829b9600b05		

log-2017-06-28_7.txt	Flight Logs	controller_android	extract_logical
SHA-256:	92bf06f4562f0e75e1d90488b722907eb275b5f01390d6a79f50f90e95cf8884		
SHA-1:	ee484996d25d0a380afadadfd6d60985028d6990		

log-2017-06-28_8.txt	Flight Logs	controller_android	extract_logical
SHA-256:	adc5a2c1d5453ab54d8a4f4e3831519f10f0d4b1ced9f71921b63e2d2cb55913		
SHA-1:	97d7ffa73e92253cc8504789e633e35618fbf58e		

log-2017-06-28_9.txt	Flight Logs	controller_android	extract_logical
SHA-256:	e349c5b9818c972494af4c7875911994e0c5d41cbd7cec14e775b86e716bfcad		
SHA-1:	52247f98480f361b27d8af8239bbccd005695b79		

log-2017-06-29.txt	Flight Logs	controller_android	extract_logical
SHA-256:	16ced7562bca75a5b79126d3f39c56737fcb01c9578af6cef39df431f3bdbfaa		
SHA-1:	0288b42b97301984d086ee75a19d55838a2512b8		

log-2017-06-29_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	459d9a0f9dc13e0c636e261721f0c3facf637759f033ab91aa3ce37ffa7550de		
SHA-1:	7ac45b648b6e7e6c364a283d474d325f1c46fb86		

log-2017-06-29_10.txt	Flight Logs	controller_android	extract_logical
SHA-256:	4841ef2585485915421bbf48b7dcae0fac7658c6fe18b6a9d7fba038e749c218		
SHA-1:	bd61e6e065cd8acaff48d138b135c18fd13faf95		

log-2017-06-29_11.txt	Flight Logs	controller_android	extract_logical
SHA-256:	2c12da720500cf38c7ba59b3eca019ec9928a6efaf37fc86207034e1d9734728		
SHA-1:	71232ec8e45799c6520ada5b6276ea0de95e0c74		

log-2017-06-29_12.txt	Flight Logs	controller_android	extract_logical
SHA-256:	78545e3586585a65f59f82e494aed68987bed09952daa0fffb5d89cde8080149		
SHA-1:	7fd17462348ccdf275832e84d681c8b011ac034c		

log-2017-06-29_13.txt	Flight Logs	controller_android	extract_logical
SHA-256:	5a61964fd80d4d666cfddec1c21d52e80d85c6704d4fb7d4968e5fc397dcf35c		
SHA-1:	e82053a17e9b9b578b36c98f93b5b15e9c2e0970		

log-2017-06-29_14.txt	Flight Logs	controller_android	extract_logical
SHA-256:	cbc7648fcd644bee43d832b9f0d5f053c00af1d45e59428e2bfbb6d656c7f305		
SHA-1:	5ed624b181be5819dd880e52d2f8af8be4eb0fbf		

log-2017-06-29_15.txt	Flight Logs	controller_android	extract_logical
SHA-256:	acd3d67fb656abfa5a9843d91cadd17cb7ca542d705a188e045a53254c1c3216		
SHA-1:	dd5dffffd40d1db8fca1679f772692642aa3c5f41		

log-2017-06-29_16.txt	Flight Logs	controller_android	extract_logical
SHA-256:	38afd509f815603917f424c2ea08383644020a78c74e0f5d1e0d8b85fc2c42c8		
SHA-1:	f3ea949033621cba251f49d95f43b9c2efe7a59c		

log-2017-06-29_17.txt	Flight Logs	controller_android	extract_logical
SHA-256:	fde91eaf607e6ffcc3336e2a7a9888282bf4f7956e7460d71929e41e625c71d00		
SHA-1:	20e11fdab24383c69c06804b6dd487d7f3d79b74		

log-2017-06-29_18.txt	Flight Logs	controller_android	extract_logical
SHA-256:	3beb4cba96c0d7a5f11faea5d4b222a9143be9fe11a16e7b9bd65fa53cd57320		
SHA-1:	5c38c064c9acb12fdb97ae945822c05e970773d2		

log-2017-06-29_19.txt	Flight Logs	controller_android	extract_logical
SHA-256:	0db709abc4279ce4039511765569e99f334ce50b21d026370a539ef5e8c0e828		
SHA-1:	5bba0bb1f018593bb460d198bcc36216f0349942		

log-2017-06-29_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	252428d67869bc14bfdbc257ecc35847b9897e8ef6d1ee45fffa7fd3e72c5c12		
SHA-1:	405c18d7691dd16244eb1a692291642f55725206		

log-2017-06-29_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	cd1520934d79d4c5db3094eaa2aa00d8de03445c808dbd64dab76817a3ceeaed		
SHA-1:	f95deb0d7839c43a0acd9896e53ad1c5b73495aa		

log-2017-06-29_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c4809ff32c480fe8f62de300af54154631cf2be71f72ddc963f58d3677adc4ad		
SHA-1:	273089627708b767f2ddbdc29049062ebefc00ef		

log-2017-06-29_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	a8794cc4ea7c3341363957e9fc470b68a78b24011eebb19e4d3debeaaaba24e		
SHA-1:	309fbc4d187288c414ed50c2e4109be87ae97093		

log-2017-06-29_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c1993f41418f813cba96b185ae26c7464e3c6800db3b28e8fbab443ca3b14c9d		
SHA-1:	33ce56f2212eae415b0cff3784de83da200e8422		

log-2017-06-29_7.txt	Flight Logs	controller_android	extract_logical
SHA-256:	591dc560a5accb90a23406bb8bb32df1d9966b17f90702c0f77c2a24c5db17ef		
SHA-1:	c58d37beb5bf30955a68f0221aaad07d5264b1ed		

log-2017-06-29_8.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9e327afb4e24abe118f5a071ee0504d6f22ecd8798feccab15c78277ee2b1b2e		
SHA-1:	257ab337fd3f4077e5852a3be3fc5ea5c2c6d774		

log-2017-06-29_9.txt	Flight Logs	controller_android	extract_logical
SHA-256:	6be59b5d89421357b65b92c14b899e0fb4d7d7febdf210552a77e36684507a5		
SHA-1:	0da050d0b3f1e4f9afb2fd289a9477b55e32cc3f		

DJIFlightRecord_2017-06-29_[12-39-25].txt	Flight Records	controller_android	extract_logical
SHA-256:	7f1d54d681ecfae94c806aafe509efad3e12ae59f482c6965431ef6ee26f9c08		
SHA-1:	713821a5cc3bfe8fd48dbe3d32a12dcd85d63e8b		

DJIFlightRecord_2017-06-29_[12-52-14].txt	Flight Records	controller_android	extract_logical
SHA-256:	cfae0772f409fe023ca42b4b5c6f55d193f6cfd6f66eb87bb295b04d1b22d3f8		
SHA-1:	8fed613ba30774488fde2011f85c6391e777984b		

thumb_49cbe7a3baed60c7_1498761563000.jpg	Images	controller_android	extract_logical
SHA-256:	e023b8e5c16746f8faa0a4efa6ca68adfee3890f00d9dc72b92e605953c9e09e		
SHA-1:	4735feb48fe7c78049f52cbd2e7211778cc2cce3		

thumb_7d5a8622435fabd_1498761669000.jpg	Images	controller_android	extract_logical
SHA-256:	0a6e6ea594e120f3a839b3da062b398ea715aa992e61926523085c4bf80ad74e		
SHA-1:	667896609f69088fab2432482c645cf0428f3e86		

DJI_0001.JPG	Images	drone_sd	extract_physical
SHA-256:	ca120108a8b18e5deb186db4ff0a0783f13be9ec340a609529c550cd565b3900		
SHA-1:	2d0d18e92731266a95383b340e929e864f262a22		

DJI_0004.JPG	Images	drone_sd	extract_physical
SHA-256:	156179a4c036f4ff09313fa145042228cdcee0792ea1687dd2bfb714a472a7ea		
SHA-1:	036116b50e76e778c28d55bccd43ffc86e71c70a		

DJI_0007.JPG	Images	drone_sd	extract_physical
SHA-256:	b6ccdc3903027f542d8fd0379c4f1001f994c638a5d5030195bbc8a7a14468fc		
SHA-1:	2f151e5af1d671012da8174f2722f1a073a0a1ac		

DJI_0008.JPG	Images	drone_sd	extract_physical
SHA-256:	ba6f37f102e847211489b5371fdb817329d8555f566a132e310763588a20b20e		
SHA-1:	0df47b41bc76c63298d8170188e7628bc9b68fea		

DJI_0010.JPG	Images	drone_sd	extract_physical
SHA-256:	01d51be119db0b02d8d7b7e31a01b115411c26dfe177eb66b5dfc3de6d60868c		
SHA-1:	dca59ae3bed017f16d4bed0f2ba8ea42f5b487ec		

DJI_0012.JPG	Images	drone_sd	extract_physical
SHA-256:	aa97f908554705972fbb0362d863056286f863d4c5b17bba34190f4440f90f8b		
SHA-1:	0248de6ae99d250e905beed71f78f9f3a66ea3bb		

DJI_0013.JPG	Images	drone_sd	extract_physical
SHA-256:	079838ccc4ed59bd93bc6aaa74a337a092063749764e9ee9326df3b3f769107d		
SHA-1:	7ffab19a4dc11ff1372acf0f6511643fbb1e5f61		

DJI_0014.JPG	Images	drone_sd	extract_physical
SHA-256:	e00139d2a43b8ae05712f06bc14e6a0f82e64869b9c182f61c61afb888109422		
SHA-1:	f399030b3368c679c3d257a4068d38cf898dda50		

DJI_0015.JPG	Images	drone_sd	extract_physical
SHA-256:	0182321ece95430b6c9dbf7e27a49956dc9761c24f5cc9b03be0a9c47ad25c6d		
SHA-1:	bb27a028481511218f1b2ad4a7ddbe6fe376003a		

DJI_0016.JPG	Images	drone_sd	extract_physical
SHA-256:	74c0691b437ec9e3911cbddf5de1b6654c4f80a36e540b128dbb02f993df566e		
SHA-1:	f67054ecf668b105b931caf07e556d4246334fbc		

DJI_0017.JPG	Images	drone_sd	extract_physical
SHA-256:	f3ebdeef41ffd3b5ccc62ec7468f6f9b7d86a541bda848ecb4dd9c162f4e49e5		
SHA-1:	8f53aa4171ad2815726870c992ae09326721d426		

DJI_0018.JPG	Images	drone_sd	extract_physical
SHA-256:	54922dafa9933d3fe0fb2ae7f657b3fb1d41c9516334e19e6a1b486fd0190993		
SHA-1:	811db60956180dc3e4441a3ef10033ef35048969		

DJI_0019.JPG	Images	drone_sd	extract_physical
SHA-256:	fbc22eb8455bd1f350c4a9fc9df97770072fcc727a354d2434dc1550b0b5b3bd		
SHA-1:	088d08ef0b0b6f0a88f77d386e9251d5ff294fc7		

DJI_0020.JPG	Images	drone_sd	extract_physical
SHA-256:	8d90018ed19e66acde8b90016826607c5416c0e38414b5e567e6b19d7815aa57		
SHA-1:	6bec9dd0cbda94a0e0e9539e0449eaaa16eebc21		

DJI_0021.JPG	Images	drone_sd	extract_physical
SHA-256:	9041aae2121b0393c0d4c0ddf3e2f15b4f0df5249b4179431dd22c46046b416b		
SHA-1:	1523175ac768e5b769a5dfc45855d14a4df8ca83		

DJI_0022.JPG	Images	drone_sd	extract_physical
SHA-256:	4169ef410cc02adc6c5eb68509a75b3da17608efc19b0ab98e77813c9cd648a0		
SHA-1:	b6627886a03a9692f5c878303c7918c3025796e7		

DJI_0023.JPG	Images	drone_sd	extract_physical
SHA-256:	beb7b3173644b5a8245f1ae715eda90da3639da617a4922b5afd35416f36fef8		
SHA-1:	faee1b8a3958c1d6386560eb56290656d0ad1f42		

2017_06_29_12_41_46.info	Videos	controller_android	extract_logical
SHA-256:	9da5fd3a57eba8acf6c7c94462b7edb23eeb78287511457c684373c955f935ba		
SHA-1:	2820b3454edfd2ba62bd1483c8ae9cd85ccfcfb8		

2017_06_29_12_41_46.mp4	Videos	controller_android	extract_logical
SHA-256:	f0881991aa4fb38512e1902a5d553dd1c8e25715158428ade1d15db874064925		
SHA-1:	6cf3f853e9c399ed7daa11447dd95c7cbfe5240d		

DJI_0002.MOV	Videos	drone_sd	extract_physical
SHA-256:	8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd02663ee668d6		
SHA-1:	3f043b78e31e45a4c0ea5687368c0f54741710bf		

DJI_0002_1.MOV	Videos	drone_sd	extract_physical
SHA-256:	b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8		
SHA-1:	4c6cc6c4e96836f07cce15495749cd62e69502bf		

DJI_0003.MOV	Videos	drone_sd	extract_physical
SHA-256:	5833218757f0835eb44ffaef533672ec0259301d097d018a6502befc59bee2f3		
SHA-1:	965cabb36ce1a935353074cd7e8937ab71dc0233		

DJI_0003_1.MOV	Videos	drone_sd	extract_physical
SHA-256:	df988a711915fffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48		
SHA-1:	3b64c1648ddc6454e2b295bf94dbd09ea4431795		

DJI_0005.MOV	Videos	drone_sd	extract_physical
SHA-256:	e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff		
SHA-1:	c6c2b79d523e66011fa2314a8ae9c01435bcf379		

DJI_0005_1.MOV	Videos	drone_sd	extract_physical
SHA-256:	5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6		
SHA-1:	60cd27bc08a8d9422b62fd032ba90fad8feb8371		

DJI_0006.MOV	Videos	drone_sd	extract_physical
SHA-256:	96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656		
SHA-1:	074aeaa3b24bc092aa482529ee96ab9010b4b6f		

DJI_0006_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514			
SHA-1:	39d075c2bac77ba15ed5c2d919033066d33d3dd3			

DJI_0009.MOV		Videos	drone_sd	extract_physical
SHA-256:	564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043			
SHA-1:	3fd13355ef8de7d3de68dcfc9933bd5f5e1b1009			

DJI_0009_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff			
SHA-1:	02c0a75b2755e5dc7e39e505b9b6f5a36656911d			

DJI_0011.MOV		Videos	drone_sd	extract_physical
SHA-256:	2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f			
SHA-1:	cec990bad5160cfabf5f32b139d2182da0d80a24			

DJI_0011_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee			
SHA-1:	c8d7b3ea1b3fc203da815aa6186a48991ff307bd			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	7	0	0
Flight Logs	40	0	0
Flight Records	2	2	2
Images	2	2	0
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	17	17	0
Videos	12	12	0

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	11	22	11

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 11:04:08 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df006_external.001

2026-06-18 11:04:08 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df006_external.001

2026-06-18 11:04:08 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df006_internal.001

2026-06-18 11:04:16 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 63 df006_internal.001

2026-06-18 11:05:30 parser_android v- rc=n/a

cmd: DF006_flight.zip C:\Users\Floris\Documents\dfdof_output\ft-df006-2017-android\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-18 11:05:35 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796677
out: DJI_0001.JPG

2026-06-18 11:05:35 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796678
out: DJI_0002.MOV

2026-06-18 11:05:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796679
out: DJI_0003.MOV

2026-06-18 11:05:59 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796680
out: DJI_0004.JPG

2026-06-18 11:06:07 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796681
out: DJI_0005.MOV

2026-06-18 11:06:28 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796682
out: DJI_0006.MOV

2026-06-18 11:06:40 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796683
out: DJI_0007.JPG

2026-06-18 11:06:40 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796684
out: DJI_0008.JPG

2026-06-18 11:06:42 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796685
out: DJI_0009.MOV

2026-06-18 11:06:45 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796686
out: DJI_0010.JPG

2026-06-18 11:06:52 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796687

out: DJI_0011.MOV

2026-06-18 11:07:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796688

out: DJI_0012.JPG

2026-06-18 11:07:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796689

out: DJI_0013.JPG

2026-06-18 11:07:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796690

out: DJI_0014.JPG

2026-06-18 11:07:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796691

out: DJI_0015.JPG

2026-06-18 11:07:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796692

out: DJI_0016.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796693

out: DJI_0017.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796694

out: DJI_0018.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796695

out: DJI_0019.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796696

out: DJI_0020.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796697

out: DJI_0021.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796698

out: DJI_0022.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796699

out: DJI_0023.JPG

2026-06-18 11:07:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044485

out: DJI_0002_1.MOV

2026-06-18 11:07:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044486

out: DJI_0003_1.MOV

2026-06-18 11:07:04 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044487

out: DJI_0005_1.MOV

2026-06-18 11:07:07 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044488

out: DJI_0006_1.MOV

2026-06-18 11:07:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044489

out: DJI_0009_1.MOV

2026-06-18 11:07:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044490

out: DJI_0011_1.MOV

2026-06-18 11:07:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 5

out: FLY000.DAT

2026-06-18 11:07:10 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 6

out: FLY001.DAT

2026-06-18 11:07:10 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 8

out: FLY002.DAT

2026-06-18 11:07:10 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 9

out: FLY003.DAT

2026-06-18 11:07:11 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 10

out: FLY004.DAT

2026-06-18 11:07:11 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 11

out: FLY005.DAT

2026-06-18 11:07:11 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 12

out: FLY006.DAT

2026-06-18 11:07:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 13

out: FLY007.DAT

2026-06-18 11:07:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 14

out: FLY008.DAT

2026-06-18 11:07:13 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 15

out: FLY009.DAT

2026-06-18 11:07:13 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 16

out: FLY010.DAT

2026-06-18 11:07:15 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTLogToCSVtool.exe DJIFlightRecord_2017-06-29_[12-39-25].txt DJIFlightRecord_2017-06-29_[12-39-25].csv

out: DJIFlightRecord_2017-06-29_[12-39-25].csv

2026-06-18 11:07:16 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTLogToCSVtool.exe DJIFlightRecord_2017-06-29_[12-52-14].txt DJIFlightRecord_2017-06-29_[12-52-14].csv

out: DJIFlightRecord_2017-06-29_[12-52-14].csv

2026-06-18 11:07:16 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_49cbe7a3baed60c7_1498761563000.jpg

out: thumb_49cbe7a3baed60c7_1498761563000_exif.json

2026-06-18 11:07:16 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_7d5a8622435fafbd_1498761669000.jpg
out: thumb_7d5a8622435fafbd_1498761669000_exif.json

2026-06-18 11:07:17 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_12_41_46.mp4
out: 2017_06_29_12_41_46_exif.json

2026-06-18 11:07:17 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.JPG
out: DJI_0001_exif.json

2026-06-18 11:07:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.JPG
out: DJI_0004_exif.json

2026-06-18 11:07:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.JPG
out: DJI_0007_exif.json

2026-06-18 11:07:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.JPG
out: DJI_0008_exif.json

2026-06-18 11:07:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0010.JPG
out: DJI_0010_exif.json

2026-06-18 11:07:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012.JPG
out: DJI_0012_exif.json

2026-06-18 11:07:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0013.JPG
out: DJI_0013_exif.json

2026-06-18 11:07:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0014.JPG
out: DJI_0014_exif.json

2026-06-18 11:07:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0015.JPG
out: DJI_0015_exif.json

2026-06-18 11:07:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0016.JPG
out: DJI_0016_exif.json

2026-06-18 11:07:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0017.JPG
out: DJI_0017_exif.json

2026-06-18 11:07:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0018.JPG
out: DJI_0018_exif.json

2026-06-18 11:07:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0019.JPG
out: DJI_0019_exif.json

2026-06-18 11:07:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0020.JPG

out: DJI_0020_exif.json

2026-06-18 11:07:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0021.JPG

out: DJI_0021_exif.json

2026-06-18 11:07:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0022.JPG

out: DJI_0022_exif.json

2026-06-18 11:07:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0023.JPG

out: DJI_0023_exif.json

2026-06-18 11:07:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MOV

out: DJI_0002_exif.json

2026-06-18 11:07:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002_1.MOV

out: DJI_0002_1_exif.json

2026-06-18 11:07:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV

out: DJI_0003_exif.json

2026-06-18 11:07:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003_1.MOV

out: DJI_0003_1_exif.json

2026-06-18 11:07:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MOV

out: DJI_0005_exif.json

2026-06-18 11:07:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005_1.MOV

out: DJI_0005_1_exif.json

2026-06-18 11:07:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.MOV

out: DJI_0006_exif.json

2026-06-18 11:07:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006_1.MOV

out: DJI_0006_1_exif.json

2026-06-18 11:07:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009.MOV

out: DJI_0009_exif.json

2026-06-18 11:07:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009_1.MOV

out: DJI_0009_1_exif.json

2026-06-18 11:07:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011.MOV

out: DJI_0011_exif.json

2026-06-18 11:07:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011_1.MOV

out: DJI_0011_1_exif.json

2026-06-18 11:07:54 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY000.DAT

out: FLY000.csv, FLY000.kml

2026-06-18 11:08:10 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY001.DAT

out: FLY001.csv, FLY001.kml

2026-06-18 11:08:27 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY002.DAT

out: FLY002.csv, FLY002.kml

2026-06-18 11:08:45 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003.DAT

out: FLY003.csv, FLY003.kml

2026-06-18 11:09:00 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT

out: FLY004.csv, FLY004.kml

2026-06-18 11:09:18 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT

out: FLY005.csv, FLY005.kml

2026-06-18 11:09:33 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

2026-06-18 11:09:47 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY007.DAT

out: FLY007.csv, FLY007.kml

2026-06-18 11:10:12 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008.DAT

out: FLY008.csv, FLY008.kml

2026-06-18 11:10:30 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY009.DAT

out: FLY009.csv, FLY009.kml

2026-06-18 11:10:45 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY010.DAT

out: FLY010.csv, FLY010.kml

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.